

Requirement Analysis Phase

Solution Requirements (Functional & Non-functional)

Date	October 3, 2026
Team ID	SWTID-2026-7210
Project Name	Script Controlled ACL: Restrict Record Access Based On Field Value
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Access Rule Configuration	• UI dashboard to create & manage custom rules • Rule storage hook configuration framework • Versioning support for dynamic security scripts
FR-2	Field-Value Interception Engine	• Database query interceptor layer hooks • Real-time contextual object field extractor • Conditional triggers based on data updates
FR-3	Script Execution Framework	• Secure sandboxed script execution engine • Context evaluation (User roles + specific field parameters) • Fallback default gate blocks on logic crashes
FR-4	Audit & Compliance Logging	• Secure audit trail compiler matrix logs • Multi-tier warning flags for illegal row queries • Exportable monitoring dashboards for internal reviews

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Clean web portal workspace enabling security managers to author clear code logic scripts with built-in auto-complete alerts and validation testers.
NFR-2	Security	Sandboxed logic interceptors running under rigorous query limits to prevent cross-site injections, database breaches, or unauthorized row escalations.
NFR-3	Reliability	99.99% operational precision. In case of unexpected rule scripting runtime evaluation errors, the system enforces a strict 'Default Deny' fallback.
NFR-4	Performance	High-speed latency constraints executing context evaluation workflows under 50 milliseconds per row query path to bypass rendering bottlenecks.
NFR-5	Availability	Continuous 99.9% middleware layer availability ensuring active dynamic rule checking instances are constantly operating without downtown hurdles.
NFR-6	Scalability	Horizontal microservices node scaling clusters to easily intercept, process, and map security restrictions across thousands of active multi-tenant streams.